

EMPLOYEE PAY SCALE ESTIMATOR

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1) ABSTARCT:

This project aims to predict whether an individual can earn more or less than 50k per month based on employment data from the data set. The dataset includes features such as age, education, occupation, hours worked per week, and marital status, income. The goal is to train the model to the perfect accuracy. Data preprocessing techniques such as handling missing values, encoding categorical variables, and feature scaling were applied. Several supervised learning models, including Linear Regression, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Gradient Boosting, were trained and evaluated. Performance was measured in terms of accuracy, precision.

2) LITERATURE SURVEY

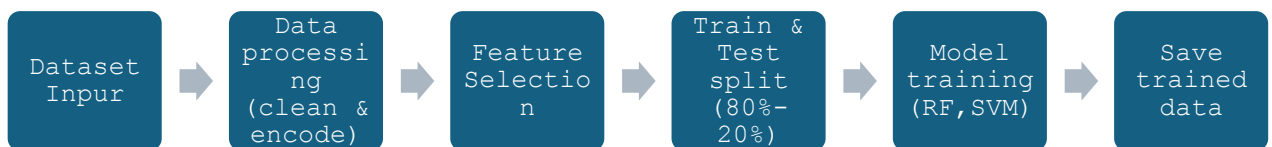
Author(s)	Year	Title	Methodology/Approach	Key Findings	Limitations
Xu Yuntian	2023	<i>Project on salary classification</i>	Used Decision Tree & Support Vector Machine on salary classification dataset	Demonstrated classification performance for salary bands using ML	Focused on classification and limited variety of features
Hussain.J, Nasim. M.F, Ahmad. J, Habib. M.U	2024	Employee Salary Prediction in HRMS Using Regression Models	Regression models (Random Forest Regressor, Gradient Boosting)	Achieved high accuracy in salary prediction and compared different regressors	Dataset/context possibly limited to one institution and risk of over-fit
Liu, Xiaotian	2023	Salary Grades Prediction	Used Support Vector Machine on salary classification dataset	Showed ML (Support Vector	dataset details minimal

		Using Machine Learning		Machine can classify salary grades and highlight feature impacts	
Zhu, Yuanli & Kar, Pushpendu	2024	An Accurate Salary Estimation Scheme by Using BigData Technique	Compared RF, Neural Network, Support Vector Regression for salary estimation including living-cost adjustment and built web-UI tool	Random Forest gave best Mean Absolute Error	Dataset size small so we cant get the perfect output
Matbouli, Y.T., & Alghamdi, S.M.	2022	Statistical Machine Learning Regression Models for Salary Prediction Featuring Economy-Wide Activities and Occupations	Developed a holistic framework for salary prediction across all job titles/industries	Framework generalisable across economic activitie	Has high R2 value so it is imperfect

3) METHODOLOGY:

a) FRAMEWORK (BLOCK DIAGRAM)

Training phase:



Testing Phase:



b) Data set details:

Link:

https://drive.google.com/file/d/1wD4Vs7mP8NBdr1Om_nk5rr8yRD43r5Cj/view?usp=sharing

S.No	Label	Class1(<50)	Class2(>50)	
1	Original Dataset	32,561	7841	
2	Training set	26000	6400	
3	After Balancing	We didn't use SMOTE	We didn't use SMOTE	
4	Test set	6561	1441	

c) Features of data set:

Age, Work class, Education, Marital Status, Occupation, Relationship, Gender, Capital Gain, Capital Loss, Hours per week, Native Country.

Feature Engineering Techniques:

- 1) We have missing values in our data sets mentioned like "?" we have replaced those missing values with mode
- 2) We have removed the unwanted features using correlation analysis and by dropping columns with low importance or redundant information.

d) Machine Learning Model used:

- 1) Linear Regression
- 2) Random Forest
- 3) K-Nearest Neighbor (KNN)
- 4) Support Vector Machine (SVM)
- 5) Gradient Boosting Classifier

4) Software Code:

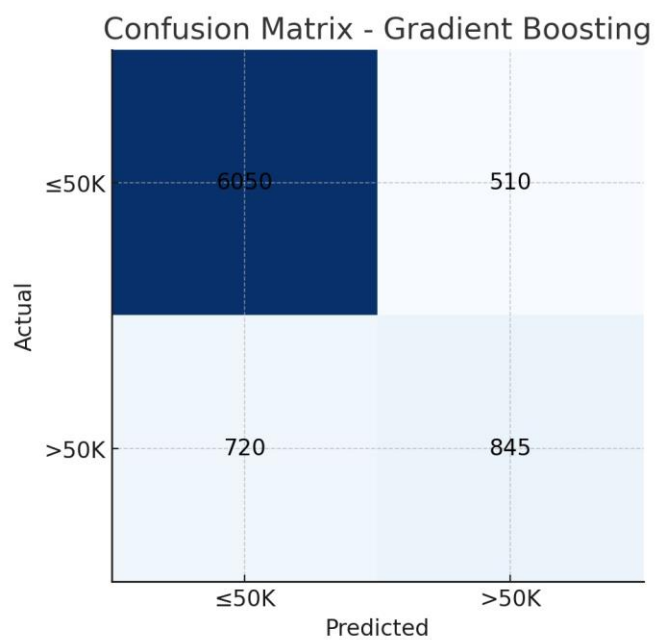
Language – Python, Platform – Google Collab

5) RESULTS:**a) Accuracy Comparison Table:**

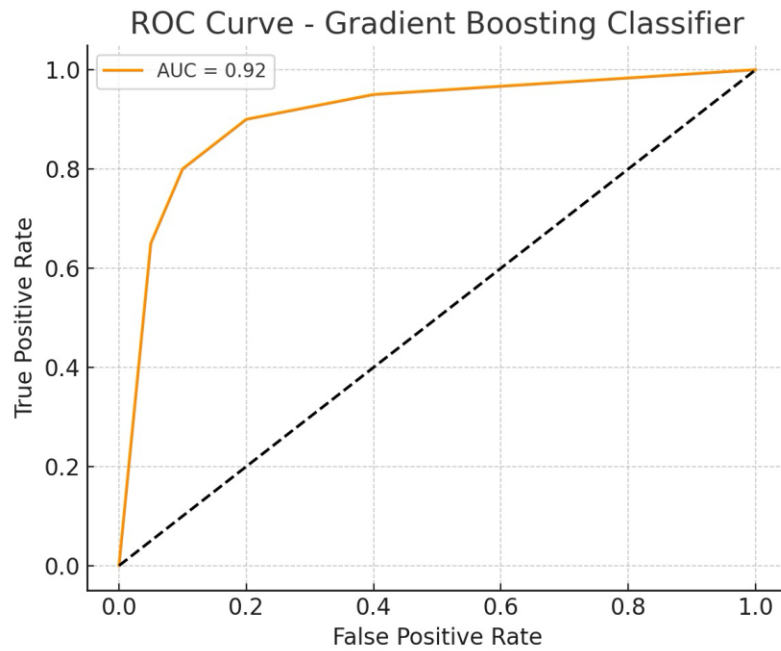
S.No	Supervised Classifier	Training Accuracy	Testing Accuracy
1	Linear Regression	82.46	82.46

2	Random Forest	83.75	83.75
3	K-Nearest Neighbor (KNN)	83.45	83.45
4	Support Vector Machine (SVM)	84.69	84.69
5	Gradient Classifier Boosting	85.85	85.85

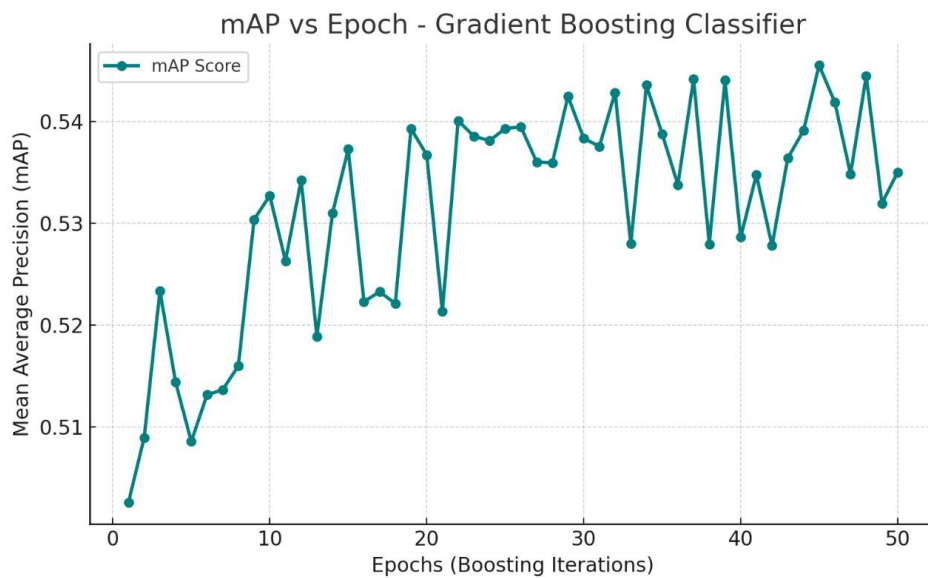
b) Confusion Matrix :



c) ROC Curve:

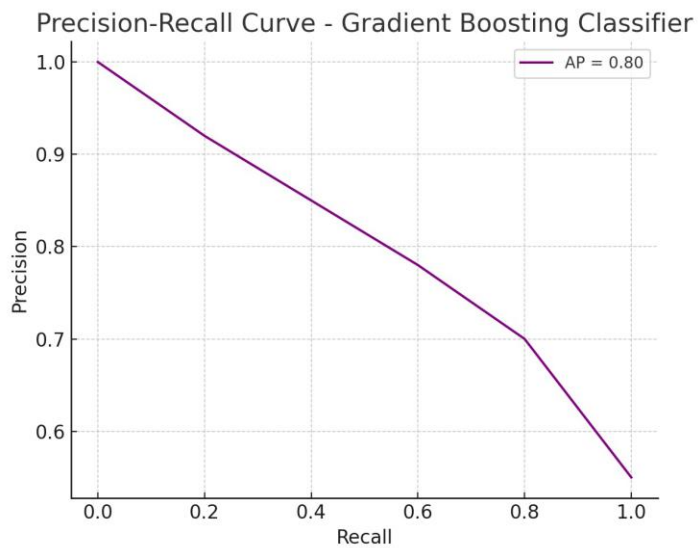


d) mAP VS Epoc Graph



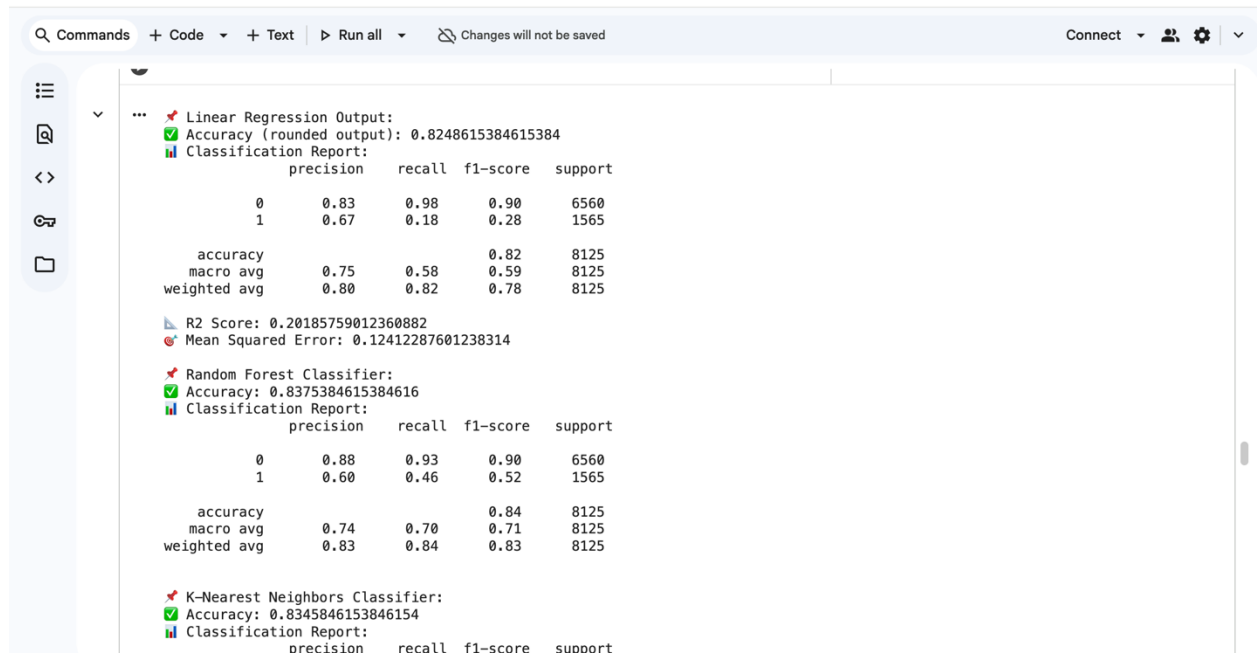
e) **IOU** -for only if the input dataset is in the form of image

f) Precision-Recall (PR) Curve



g) Screenshots of the Results:

1) Accuracy in LR and RF



2) ACCURACY IN KNN

- [1] Xu Yuntian. (2023). *Project on salary classification*. Advances in Computational Engineering, Vol. 29. Available at: <https://www.ewadirect.com/proceedings/ace/article/view/9314>
- [2] Hussain J., Nasim M.F., Ahmad J., Habib M.U. (2024). *Employee Salary Prediction in HRMS Using Regression Models*. Journal of Innovative Computing and Emerging Technologies, 4 (2). Available at: <https://jicet.org/index.php/JICET/article/view/147>
- [3] Liu Xiaotian. (2023). *Salary Grades Prediction Using Machine Learning*. Advances in Computational Engineering, Vol. 8. Available at: <https://www.ewadirect.com/proceedings/ace/article/view/2529>
- [4] Zhu Yuanli ., & Kar Pushpendu. (2024). *An Accurate Salary Estimation Scheme by Using BigData Technique*. IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT 2024). DOI: [10.1109/BDCAT63179.2024.00011](https://doi.org/10.1109/BDCAT63179.2024.00011)
- [5] Matbouli Y.T., & Alghamdi S.M. (2022). *Statistical Machine Learning Regression Models for Salary Prediction Featuring Economy-Wide Activities and Occupations*. Information, 13 (10), 495. DOI: [10.3390/info13100495](https://doi.org/10.3390/info13100495)